

Introduction to Pulmonology

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Introduction to Pulmonology – Respiratory Paediatrics

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Outline

What pulmonology is · Embryology of the respiratory system · Developmental phases & surfactant · The respiratory tract as a continuum · Upper vs lower tract · Trachea, bronchi and lungs · Pathophysiology · Classification · Evaluation: history, examination, investigation

What is pulmonology?

Pulmonology is the field of medicine that focuses specifically on **diagnosing and treating disorders of the respiratory system**. It is also known as **respirology, respiratory medicine or chest medicine**.

Embryology of the respiratory system

Development of the respiratory system from the foregut

In very early development, the embryo has a simple tube called the **foregut**. A partition grows to split this tube into two channels:

- One going to the stomach – the **oesophagus**
- One going to the lungs – the **trachea**

Everything that becomes the airway – **larynx, trachea, bronchi, alveoli** – grows out from that single anterior split. The respiratory system grows as an **outgrowth from the ventral (front) wall of the foregut in week 4**.

Tissue origins

- The **epithelial lining** of everything from the larynx down to the alveoli is **endodermal** (from the endoderm)
- The **cartilage and muscle** wrapping around the airways are **mesodermal** (from the mesoderm)

The oesophagotracheal septum

The **oesophagotracheal septum** is the wall of tissue that grows to **separate the respiratory bud from the oesophagus (foregut)**. It does this in the **4th week**.

Failure of this septum to form properly causes tracheo-oesophageal fistula (TOF)
– a connection between the airway and the food pipe, which is a **surgical emergency in newborns**.

The septum divides the foregut into:

- The **respiratory diverticulum anteriorly** (becomes the lungs)
- The **oesophagus posteriorly**

The two remain **connected at one point – through the larynx**.

The **larynx** is formed from tissue of the **4th and 6th branchial arches**.

Lung bud asymmetry

Then the lung buds diverge:

- The **left bud** forms **2 main bronchi and 2 lobes**
- The **right bud** forms **3 main bronchi and 3 lobes**

This asymmetry is not random – it is determined from this embryological stage, and it is **why the adult right lung is slightly larger**.

Developmental phases & surfactant production

Two key phases of lung maturation:

- **Pseudoglandular phase (5-17 weeks)** – the lung looks **gland-like** under the microscope. Airways are forming but there is **no gas exchange yet** – hence “false gland”
- **Canalicular phase (15-25 weeks)** – the airways begin to **widen and vascularize**. The cells lining the bronchioles **thin out and flatten into type I alveolar epithelial cells** (the actual gas-exchange cells). Also during this period, **type II alveolar epithelial cells begin producing surfactant** – a phospholipid coat laid over the alveolar membranes before birth

What surfactant does

Surfactant's job is to **reduce surface tension at the air-blood capillary interface**. Without it, the natural surface tension of the fluid lining the alveoli would cause them to **collapse inward during expiration**. Surfactant keeps the alveoli open so the next breath comes in easily.

This is exactly why **premature infants** (born before sufficient surfactant is produced) develop **RDS (respiratory distress syndrome)** – their alveoli keep collapsing and breathing becomes exhausting work.

Postnatal lung growth

After birth, the lungs grow by **increasing the number** of respiratory bronchioles and alveoli – **not by making existing alveoli bigger**. Lung growth is about **multiplication, not enlargement**.

The respiratory tract as a continuous highway

The respiratory tract from nasal cavity to alveoli

The respiratory tract is **one single continuous tract** that starts at the nose and ends deep in the alveoli – it flows as **one connected mucosal surface**.

This matters clinically because an **infection that starts at the top of the road can travel downward** along it. There are no real anatomical “stop signs” between the upper and lower tract.

The road is divided into two sections at a practical landmark – the **larynx**. Everything **above it is the upper respiratory tract**; everything **below is the lower respiratory tract**.

The full continuum:

Nasal cavity -> Pharynx -> Epiglottis -> Larynx -> Trachea -> Bronchi -> Bronchioles -> Alveolar sacs

The word “**continuum**” tells you why a child with a runny nose can progress to bronchitis or pneumonia – the infection simply travels down the same road.

Upper vs lower respiratory tract

Upper respiratory tract	Lower respiratory tract
Nose	Trachea
Paranasal sinuses	Bronchi
Pharynx	Bronchioles
	Alveoli

The **upper** structures can be examined from the outside or with a simple light. The **lower** structures are deeper, require instruments to examine, and are where the serious diseases – **pneumonia, asthma, bronchiolitis** – tend to live.

The trachea

Position of the trachea – cricoid at C6, bifurcation at the sternal angle

- **Where it starts** – at the **lower border of the cricoid cartilage at C6**
- **Where it ends** – the trachea **bifurcates at the sternal angle of Louis (T4/T5)** to form the right and left main bronchi
- **Dimensions** – about **4½ inches long, nearly 1 inch in diameter**
- **Lining** – **ciliated columnar epithelium with goblet cells**. The cilia beat upward to sweep debris out; the goblet cells produce mucus to trap pathogens – part of the respiratory tract's **first-line defence system**

The sternal angle of Louis – the surface landmark for the tracheal bifurcation

Surface anatomy of the chest

The bronchi

The carina and the differing angles of the two main bronchi

The trachea bifurcates into two main bronchi, with a **crucial asymmetry**:

- The **right main bronchus** is **wider, shorter and more vertical**, leaving at an angle of **25° to the horizontal**
- The **left main bronchus** leaves at a **steeper 45°**

Why foreign bodies go right

Because the right bronchus is **wider and more vertical** – essentially a direct continuation of the trachea – **foreign bodies and aspirated material preferentially enter the right bronchus**, particularly into the **middle and lower lobes** of the right lung.

This is **one of the most tested clinical anatomy facts in paediatrics**. If a child inhales something, **look right first**.

Each lung is **conical**, with a **blunt apex projecting above the sternal end of the first rib** – which is why a **penetrating injury at the base of the neck can still injure the lung**.

Lobes of the lungs

Lobes and fissures of the lungs

- The **right lung** is slightly larger – **three lobes** (upper, middle, lower) divided by **two fissures**: the **oblique** and the **horizontal**
- The **left lung** has only an **oblique fissure**, giving **two lobes** (upper and lower). It is smaller partly because the **heart occupies space on the left**

Bronchopulmonary segments

The clinical consequence of continuity

Because the respiratory tract is **one connected surface**, infections rarely stay confined to one region. **Downward spread along the tracheobronchial tree is common** – a child with a URTI can progress to **bronchitis, bronchiolitis or pneumonia** within days as the same pathogen moves along the mucosal highway.

Pathophysiology

The main function of the respiratory system is to **supply sufficient oxygen to meet metabolic demands and remove carbon dioxide**. This involves **ventilation, perfusion and diffusion**. **Abnormality in any one** of these mechanisms can lead to **respiratory failure**.

Classification

Respiratory tract disease is classified into:

- **Upper Respiratory Tract Infection (URI / URTI)**
- **Lower Respiratory Tract Infection (LRI / LRTI)**

The three pillars of evaluation

Evaluating a child with respiratory disease rests on three things, **in this deliberate order**:

1. **History taking**
2. **Physical examination**
3. **Laboratory investigation**

History comes first because it guides everything else – it tells you what to look for on examination and which investigations to request. Without a good history, examination and investigation become **unfocused and expensive**.

The structure of the history

Five components: **bio-data · presenting symptoms · past medical history · family history · social history**.

Bio-data: age and gender

Infant vs adult rib cage – the compliant infant chest wall

Age gives insight into anatomical maturity, physiological reserve, likely diagnosis and risk of deterioration.

Why young infants are more vulnerable (the immature respiratory system):

- **Smaller airways** – a small amount of mucus, oedema or inflammation narrows the airway significantly. Since **resistance rises dramatically as airway diameter falls**, infants obstruct faster
- **Highly compliant (floppy) chest wall** – soft, horizontal ribs mean the chest wall **collapses inward** in distress instead of supporting ventilation -> **recession/retractions and quicker respiratory fatigue**
- **Immature respiratory control / poor hypoxic reserve** – infants tolerate hypoxia poorly and can develop **apnoea**, especially neonates
- **Lower physiological reserve** – limited oxygen reserve and higher metabolic demand mean they **decompensate faster**

A 2-month-old with bronchiolitis and an 8-year-old with the same severity are **not equally stable** – the infant has far less reserve.

Age predicts the likely diagnosis:

Age	Common respiratory diseases
Neonate (<28 days)	Congenital anomalies, neonatal pneumonia, aspiration
Infant (<1 year)	Bronchiolitis, pertussis
Toddler (1-3 years)	Viral wheeze, foreign body aspiration, croup
School-age	Asthma, pneumonia
Adolescent	Asthma, atypical pneumonia

Gender – some conditions have a sex predilection; **pneumonia and bronchiolitis are more common in males**, adjusting your pre-test probability.

Presenting symptoms: the big three

Cough – characterise it: **duration**, **character** (barking, whooping, productive?), **timing** (night, morning, after feeding), **severity**, and **sputum production**.

Breathlessness – how severe, how long, and the pattern (constant or episodic, with exercise or at rest?).

Chest pain – the key task is distinguishing **respiratory** chest pain (relates to breathing – **worse on inspiration or coughing = pleuritic**) from **cardiac** chest pain.

Noisy breathing – six types you must know

Wheeze · Stridor · Snuffles · Rattles · Snoring · Grunting

A child in respiratory distress

Other associated symptoms

- **Rhinorrhoea and nasal stuffiness** -> upper tract involvement or allergic disease
- **Ear ache or discharge** -> otitis media often accompanies URTIs (short, horizontal Eustachian tube in young children)
- **Hoarseness** -> laryngeal involvement – croup, laryngitis, or a laryngeal foreign body
- **Sore throat** -> pharyngitis or tonsillitis
- **Red flags – weight loss, neck swelling, night sweats** -> raise suspicion of **tuberculosis or lymphoma**, which must never be missed

Past medical history

Tells you whether this is **new or recurring**. Five things to probe: **has it happened before?** · **has the child been hospitalised for it?** · **did symptoms follow a previous infection?** · **how did the child respond to previous treatment** (did bronchodilators or antibiotics help)? · **is there a seasonal pattern?**

A child with **recurrent wheezing that responds to bronchodilators and worsens in harmattan** is telling you the diagnosis before you examine them – this reversible, trigger-worsened pattern is classic for **bronchial asthma**.

Family history

- **Atopy** – the tendency to allergic disease (asthma, eczema, allergic rhinitis) runs strongly in families; **two atopic parents** greatly raise the child's risk
- **Chronic cough** in the family may point to **cystic fibrosis or primary ciliary dyskinesia** (genetic bases)

Social history – the environment the child lives in

- **Dust, pets, rugs and fumes** can trigger allergic and obstructive disease
- **Passive smoking** is one of the most significant environmental risk factors – it **damages airway mucosa, impairs ciliary function and increases infection susceptibility**
- **Exercise** can trigger bronchospasm in reactive airway disease
- **Overcrowding** is a major risk factor for spread of respiratory infection, especially **TB and pneumonia**

The social history is where you find the **hidden causes** that the symptoms alone won't reveal.

References

Lecture slides of Dr (Mrs) Odochi Ewurum; standard paediatric respiratory texts.